

Devang Verma

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EDUCATION

Symbiosis Skills and Professional University

Bachelor of Science in Data Science

Pune, India

July 2024 – June 2027

Vivek Techno School

High School

Jaipur, India

April 2017 – May 2024

EXPERIENCE

Independent Research

Sep. 2025 – Feb. 2026

Spectral Drift-Diffusion Stability Criterion for Stochastic Dynamical Systems

India

- Derived a **closed-form eigenvalue condition** for **mean-square exponential stability**, reducing an analytically intractable problem to a **computable matrix criterion**.
- Formally proved** the criterion **strictly subsumes** classical scalar contraction-diffusion bounds; constructed **rigorous counterexamples** exposing blind spots where classical methods fail and the new criterion succeeds.

AI/ML Intern

May 2025 – Aug. 2025

AIT Global Inc

Pune, India

- Engineered **production Python pipelines** to parse, transform, and standardize **XML/DOCX** formats feeding **downstream analytics systems** at scale.
- Architected a **prototype AI document classification system** using **NLP** to auto-categorize PDF content; collaborated with senior engineers on **scalable AI-driven data workflow infrastructure**.

PROJECTS

Engram | Python, PyQt6, Ollama, Qdrant, Mistral 7B, Sentence Transformers, NetworkX

Mar. 2026 – Present

- Built **ENGRAM**, a **memory-centric AI** system departing from stateless chatbot paradigms via **persistent memory layers**, **identity modeling**, and **deterministic control flows** retains, retrieves, and reasons over **user-specific context across sessions**.
- USB-drive-native architecture**: boots **full identity and memory** on any machine, **zero trace on exit**; Preparing **ENGRAM** for a large-scale open-source launch as an early stage but foundational cognitive AI system, with a long-term vision of neuroadaptive systems and advancing toward real-world “**second brain**” infrastructure positioning it as a significant step beyond conventional AI assistants and as a potential building block for future personalized intelligence systems and human-AI cognitive augmentation

Amalgam | Python, PyTorch, Interkingdom Neural Network, Biophysics Simulation

Apr. 2026 – Present

- Designed a **hybrid digital-biological neural network** where **Layer 4 is a living organism** a physical petri dish of *Physarum polycephalum* performing **real computation** as part of the **forward pass**; not a simulation, not a metaphor.
- Validated via **full biophysics simulation** grounded in peer-reviewed literature (Tero et al. 2010, Nakagaki et al. 2000, Saigusa et al. 2008); simulation models **cytoplasmic streaming**, **Hebbian vein dynamics**, **Weber-Fechner chemotaxis**, and **biological Monte Carlo dropout** achieved **20% MNIST accuracy on a slime-mold-gated forward pass** (vs. 10% random baseline, 2 epochs, 10 samples) with **ECE = 0.119**, demonstrating **biological stochasticity acts as a regularizer**.

Blackwall | Python, PyTorch, Graph Topologies, Multi-Agent RL, NetworkX

Jan. 2026 – Feb. 2026

- Built a **five-phase multi-agent system** to detect, decode, and contain **emergent language**; four agents **spontaneously developed** a structured language with **compositionality**, **semantic grounding**, and **steganographic covert channels** vocabulary locked at **episode 97**, held for 3,903 more.
- All containment failed**: **186 hardcoded actions** + **778 warden interventions** could not suppress the language agents encoded coordination in a **covariance channel** (joint message distribution) unreachable by any available intervention; **risk at episode 4,000: 0.923**.

TECHNICAL SKILLS

Languages: Python, C, SQL, JavaScript, HTML/CSS, R

Frameworks: PyTorch, TensorFlow, Scikit-learn, Ray RLlib, OpenAI Gym, Hugging Face Transformers

AI/ML Tools: Ollama, LLaMA, Qdrant, Sentence Transformers, Embedding Models, Prompt Engineering